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| **Fiscal Year:** | FY 2012 | **Task Last Updated:** | FY 02/13/2012 |
| **PI Name:** | Miller, Fletcher J Ph.D. | | |
| **Project Title:** | Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | COMBUSTION SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Combustion science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | UNIVERSITY | **Phone:** | 619-594-5791 |
| **Organization Name:** | San Diego State University | | |
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| **City:** | San Diego | **State:** | CA |
| **Zip Code:** | 92182-1323 | **Congressional District:** | 53 |
| **Comments:** | Previously at National Center for Space Exploration Research, NASA Glenn Research Center, until 2007. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2009 Combustion Science NNH09ZTT001N |
| **Start Date:** | 01/01/2010 | **End Date:** | 12/31/2014 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 5 | **No. of Bachelor's Degrees:** | 7 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Olson, Sandra | **Contact Phone:** | 216-433-2859 |
| **Contact Email:** | Sandra.Olson@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date is now 12/31/2014 per NSSC information (Ed., 1/22/14)<br>NOTE: End date is now 12/31/2013 per PI (Ed., 10/11/2012) | | |
| **Key Personnel Changes/Previous PI:** | During the summer of 2010 Prof. George Sidebotham from Cooper Union participated in the research as a NASA Summer Faculty Fellow at NASA Glenn Research Center. | | |
| **COI Name (Institution):** | Olson, Sandra ( NASA Glenn Research Center )<br>Wichman, Indrek ( Michigan State University ) | | |
| **Grant/Contract No.:** | NNX10AD96A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | This research will study flame spread over flammable solid materials such as plastic, paper, cloth, and hair, using a new apparatus designed to simulate microgravity flow conditions on earth. We designed and built two new Narrow Channel Apparatuses at San Diego State, and one at Michigan State, that control environmental variables such as flow velocity, gas pressure, and oxygen concentration to reproduce space craft atmospheres. Additionally, the Narrow Channel suppresses buoyant flow by having a closely spaced ceiling on the channel, so that flow conditions closely resemble those found on space craft. The objective of the research is to prove the Narrow Channel Apparatus as a viable test method for opposed flow material flammability testing. This will allow NASA to screen materials in more realistic conditions than does the current NASA test which makes no account of microgravity. The experimental research consists of test wherein we ignite various solids in the channels and measure how fast the flame spreads in various conditions. To accompany the experiments, we are developing a numerical model so that we can determine certain quantities such as heat loss to the channel ceiling that are difficult to measure. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our research encompasses flame spread and material flammability in narrow channels or gaps. We use this geometry to simulate microgravity conditions. However, many fire situations involve flames in narrow gaps. TWA Flight 800, which crashed off the coast of Canada, was brought down by a small fire spreading through some narrow gaps in the walls and ceiling near the cockpit, for instance. Many building fires involve fire spread inside walls. Fires in computers likewise can occur between the printed circuit boards. Thus, information gained from experiments about how flames spread in various atmospheric conditions in narrow gaps is useful in areas well beyond spacecraft fire safety. The numerical model we are developing will be validated against our experiments and can also be used to study flame spread in these gaps. |
| **Task Progress:** | During the first year of this grant we designed and constructed three apparatuses to study how flames spread in narrow channels under varying atmospheric conditions. The fuels planned in the tests are those that would be of interest to NASA such as paper, plastic, cloth, and hair, so that material flammability can be determined. These narrow channels, due to their ability to restrict hot air rising, simulate microgravity conditions on spacecraft where heat also does not "rise," since there is no "up" in microgravity. The three channels are all different to allow different effects to be tested. The first, the San Diego State University Narrow Channel Apparatus (SDSU NCA), allows sample materials 2 inches wide by up to 18 inches long to be burned. The channel is capable of being operated at low pressure up to 2 times atmospheric pressure and with oxygen concentrations from 85% to 21% (like on earth) to simulate different spacecraft atmospheres. An air flow over the sample simulates the spacecraft ventilation system. The airflow has a character like that found in a small gap, and therefore does not completely reproduce the flow in an open spacecraft cabin.<br><br>The second channel at SDSU can test samples the same size, but due to a moving top on the channel, it creates a flow much more like an open cabin flow that might exist in a space craft fire. It cannot, however, recreate pressure conditions on space craft. This is the first channel of its kind in the world, and it is still undergoing some trial tests.<br><br>The third channel, at Michigan State, is a much wider channel than the others so that it can test large samples up to 12 inches wide. It is not capable of reduced pressure, but can vary the flow velocity and oxygen level.<br><br>All three channels can have a variable gap height and flow velocity, and each is outfitted with a camera to record videos of the flame to determine how fast it spreads. The gap height is defined as the distance between the fuel and wall, and is approximately 1/4 inch.<br><br>During the second year of the study, velocity measurements have been made in each channel to calibrate them. Then, flame spread tests have been conducted using various paper fuels similar to those used by other investigators to see the effect of pressure, oxygen level, and flow velocity for a given gap height. Many of the conditions tested were obtained for the first time and new data were recorded. These data give a much clearer overview of flame spread in different conditions than had been available before.<br><br>Separately from the efforts above, the original Narrow Channel at NASA was used to test the effect of gap height and an optimized height was found that was used in the other tests. Also, a series of tests were conducted with trioxane as the fuel, since that was a possible fuel for microgravity tests on the International Space Station. These were the first flame spread tests known to be conducted over trioxane. |
| **Bibliography Type:** | Description: (Last Updated: 06/15/2021) |
| **Papers from Meeting Proceedings** | Pepper J, Miller F. "Characterizing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test." 2011 Fall Technical Meeting of the Western States Section of the Combustion Institute, Riverside, CA, October 16-18, 2011.<br>In: Fall Technical Meeting of the Western States Section of the Combustion Institute 2011. Livermore, Calif. : Western States Section/Combustion Institute, 2011. p. 683-694. Paper 027FR-0042. Printed by Curran Associates, Inc. ISBN: 978-1-61839-304-3. , Dec-2011 |
| **Papers from Meeting Proceedings** | Hung K, Olson S, Miller F. "Flame Spread Across Trioxane in a Narrow Channel Apparatus Simulating Microgravity Flow Conditions." 2011 Fall Technical Meeting of the Western States Section of the Combustion Institute, Riverside, CA, October 16-18, 2011.<br>In: Fall Technical Meeting of the Western States Section of the Combustion Institute 2011. Livermore, Calif. : Western States Section/Combustion Institute, 2011. p. 862-879. Paper 027FR-0178. Printed by Curran Associates, Inc. ISBN: 978-1-61839-304-3. , Dec-2011 |
| **Papers from Meeting Proceedings** | Martinez D, Miller F. "Numerical Modeling of Cellulose Combustion in a Narrow Channel Apparatus That Simulates Microgravity Flow Conditions." 2011 Fall Technical Meeting of the Western States Section of the Combustion Institute, Riverside, CA, October 16-18, 2011.<br>In: Fall Technical Meeting of the Western States Section of the Combustion Institute 2011. Livermore, Calif. : Western States Section/Combustion Institute, 2011. p. 837-853. Paper 027FR-0082. Printed by Curran Associates, Inc. ISBN: 978-1-61839-304-3. , Dec-2011 |

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| **Papers from Meeting Proceedings** | Sidebotham GW, Olson SL, Rachow P, Miller FJ, Wichman IS. "Effect of Gap Height on Thin Fuel Opposed Flow Flame Spread in the Narrow Channel Apparatus." 7th US National Technical Meeting of the Combustion Institute, Georgia Institute of Technology, Atlanta, GA, March 20-23, 2011.<br>In: 7th US National Technical Meeting of the Combustion Institute, Georgia Institute of Technology, Atlanta, GA, March 20-23, 2011. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 1746-1751. Printed by Curran Associates, Inc. ISBN: 978-1-61782-780-8 , Mar-2011 |